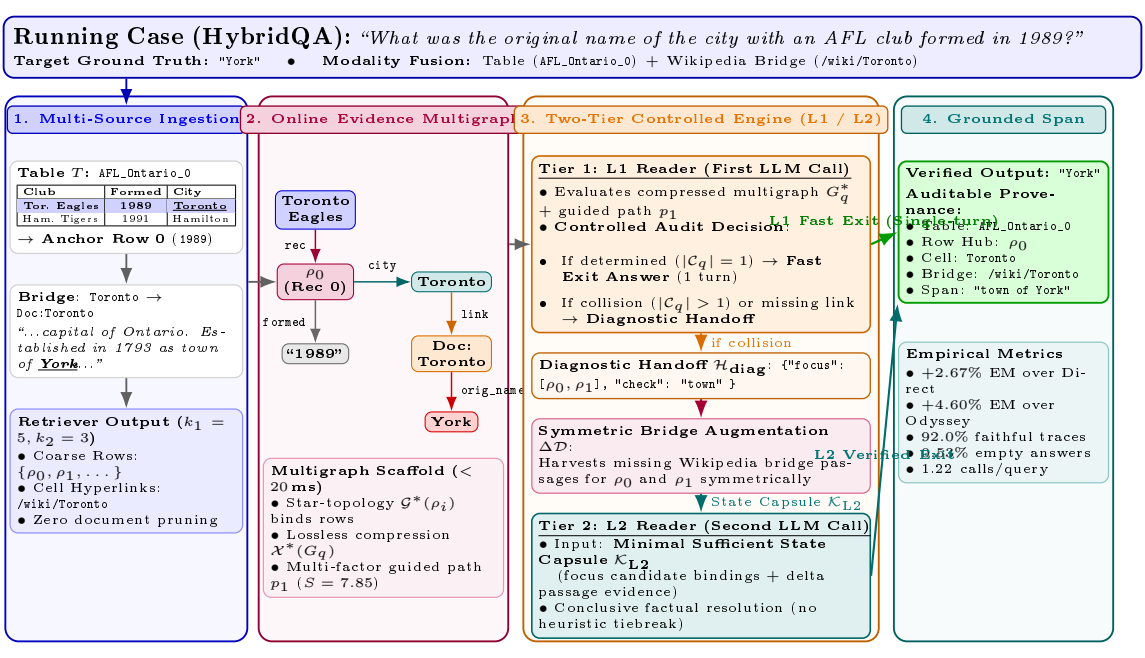

Running Case (HybridQA): "What was the original name of the city with an AFL club formed in 1989?"
Target Ground Truth: "York" • Modality Fusion: Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)
1. Multi-Source Ingestion
Table T: AFL_Ontario_0
Club | Formed | City
Tor. Eagles | 1989 | Toronto
Ham. Tigers | 1991 | Hamilton
→ Anchor Row 0 (1989)
Bridge: Toronto → Doc:Toronto
"...capital of Ontario. Established in 1793 as town of York..."
Retriever Output ($k_1 = 5, k_2 = 3$)
• Coarse Rows: $\{\rho_0, \rho_1, \dots\}$
• Cell Hyperlinks: /wiki/Toronto
• Zero document pruning
2. Online Evidence Multigraph
Toronto Eagles
rec
ρ_0 (Rec 0)
city
Toronto
formed
"1989"
link
Doc: Toronto
orig_name
York
Multigraph Scaffold (< 20 ms)
• Star-topology $\mathcal{G}^*(\rho_i)$ binds rows
• Lossless compression $\mathcal{X}^*(G_q)$
• Multi-factor guided path p_1 ($S = 7.85$)
3. Two-Tier Controlled Engine (L1 / L2)
Tier 1: L1 Reader (First LLM Call)
• Evaluates compressed multigraph G_q^* + guided path p_1
• Controlled Audit Decision:
• If determined ($|\mathcal{C}_q| = 1$) → Fast Exit Answer (1 turn)
• If collision ($|\mathcal{C}_q| > 1$) or missing link → Diagnostic Handoff
L1 Fast Exit (Single-turn)
if collision
Diagnostic Handoff $\mathcal{H}_{\text{diag}}$: {"focus": [$\rho_0, \rho_1$], "check": "town" }
Symmetric Bridge Augmentation $\Delta\mathcal{D}$:
Harvests missing Wikipedia bridge passages for ρ_0 and ρ_1 symmetrically
L2 Verified Exit
State Capsule $\mathcal{K}_{\text{L2}}$
Tier 2: L2 Reader (Second LLM Call)
• Input: Minimal Sufficient State Capsule $\mathcal{K}_{\text{L2}}$ (focus candidate bindings + delta passage evidence)
• Conclusive factual resolution (no heuristic tiebreak)
4. Grounded Span
Verified Output: "York"
Auditable Provenance:
• Table: AFL_Ontario_0
• Row Hub: ρ_0
• Cell: Toronto
• Bridge: /wiki/Toronto
• Span: "town of York"
Empirical Metrics
• +2.67% EM over Direct
• +4.60% EM over Odyssey
• 92.0% faithful traces
• 0.53% empty answers
• 1.22 calls/query